

Will Tomorrow Be Lit By LEDs?

The performance of LEDs has improved so much over time that these tiny bulbs are all set to brighten homes, light up streets, project images from mobile phones, and get involved in our lives in many more interesting ways. Governments are even challenging companies to come up with viable, sustainable, LED-based home lighting solutions, cheap enough to replace the omnipresent 60W bulb. Eco-friendly, low-power LED lamps might be the biggest step we take towards saving this planet!

■ JANANI GOPALAKRISHNAN VIKRAM



Do you remember the HP-35 calculator that Hewlett-Packard (HP) launched way back in 1972? It was a hot favourite amongst engineers, being the only pocket calculator with transcendental functions at that time. It cost \$395, and believe it or not, the price tag was so heavy because the device used light-emitting diodes (LEDs)! Today, LEDs are used even by school children for their science projects—say, an experiment to light up little bulbs using a spinning magnet. That is how common these tiny ‘digital’ lights have become.

At a time when their cost was prohibitive, LEDs were used merely as indicators in electronic devices. The fact that these bud-like lights consumed very little power, gave out very little heat and had a long life made them very popular and led manufacturers to explore various other applications. They are now used as backlights for digital displays, including liquid crystal display (LCD) flat televisions and computer monitors, sustainable lighting systems such as solar lanterns and solar street lights, as well as conventional home lighting. Even as these applications slowly gain momentum, newer ones such as automotive lighting and mobile projection are cropping up.

This quick uptake of LED technology can be attributed to the fact that the industry has been keeping pace with Haitz’s Law—a prediction by an erstwhile HP employee called Ronald Haitz that the amount of light that can be produced per diode (performance) would increase twenty-fold every decade, while the cost of that light would decrease ten-fold.

Paul Thieken, director of marketing-LED Components, Cree, stresses that over the past 40 months or so, the performance of high-power LEDs has improved drastically. “The most exciting improvements are the greatly increased light output (flux), improved efficacy (lumens-per-watt) and the colour quality and stability. These im-